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EXP.NO:3 Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree

dataset = pd.DataFrame({
    "Outlook": ["Sunny", "Sunny", "Overcast", "Rain", "Rain", "Overcast", "Sunny", "Rain"],
    "Temperature": ["Hot", "Hot", "Hot", "Mild", "Cool", "Cool", "Mild", "Mild"],
    "Humidity": ["High", "High", "High", "High", "Normal", "Normal", "Normal", "High"],
    "Wind": ["Weak", "Strong", "Weak", "Weak", "Weak", "Strong", "Strong", "Strong"],
    "Play": ["No", "No", "Yes", "Yes", "Yes", "Yes", "Yes", "No"]
})

print("Dataset")
print(dataset)

le = LabelEncoder()
for column in dataset.columns:
    dataset[column] = le.fit_transform(dataset[column])

X = dataset.iloc[:, :-1]
y = dataset.iloc[:, -1]

model = DecisionTreeClassifier(criterion="entropy", random_state=0)
model.fit(X, y)

plt.figure(figsize=(10,6))
plot_tree(
    model,
    feature_names=["Outlook", "Temperature", "Humidity", "Wind"],
    class_names=["No", "Yes"],
    filled=True,
    rounded=True
)

plt.title("Decision Tree using ID3 Algorithm")
plt.show()
```

```

sample = pd.DataFrame({
    "Outlook":["Sunny"],
    "Temperature":["Cool"],
    "Humidity":["High"],
    "Wind":["Weak"]
})

for column in sample.columns:
    encoder = LabelEncoder()
    encoder.fit(["Sunny","Rain","Overcast"] if column=="Outlook"
               else ["Hot","Mild","Cool"] if column=="Temperature"
               else ["High","Normal"] if column=="Humidity"
               else ["Weak","Strong"])
    sample[column] = encoder.transform(sample[column])

prediction = model.predict(sample)
if prediction[0] == 1:
    print("\nPrediction: Play = Yes")
else:
    print("\nPrediction: Play = No")

```

OUTPUT:

